Bank Management System

A practical guide to using the desktop application

This application is an educational simulation. Do not use it to hold or move real funds, make real lending decisions, or store sensitive real customer information.

Language: English | Format: Desktop application

1. About the application

This application is a practice environment for common banking workflows. It includes customer records, accounts, transactions, loans, interest credits, reports, staff permissions, and backups.

Records are stored in the application's local database on this computer. Changes are saved as you use the application. Administrators should still create regular backups.

- Open the desktop shortcut. If you start it from the project folder, run the main.py application file.

2. First launch and sign-in

- On the first launch, create the initial administrator account by choosing a username and password.
- A password must have at least 8 characters, including at least one letter and one digit. Keep it private.
- After setup, enter your username and password on the sign-in screen.
- Choose English from the language menu on the sign-in screen or dashboard. Your language choice is saved for the next launch.

Several incorrect sign-in attempts temporarily lock that username for the current application session. Wait briefly, then try again.

3. User roles

- Administrator (admin): can view all sections, review customer verification, approve or reject loan requests, disburse approved loans, view reports, manage staff, create or restore backups, and review transaction reversals. Administrators cannot perform routine deposits, withdrawals or transfers.
- Bank employee (teller): can manage customers and accounts, deposit, withdraw and transfer funds, handle loan applications and repayments, and preview or apply monthly interest.
- Read-only user (viewer): can view and search customers, accounts and transactions, and view reports. This role cannot change records.

If a control is missing or disabled, your role may not have permission for that action. Ask an administrator.

4. Dashboard and navigation

After sign-in, the dashboard shows an overview of customer and account activity. Use the navigation menu to open a section. When Pashto is selected, the menu moves to the right side.

- Customers: create customer records and review verification status.
- Accounts: open accounts, search records and change account status.
- Transactions: deposit, withdraw, transfer and search transaction history.
- Loans and Interest: manage loan requests and account interest.
- Reports: view summaries, check account balances and export files.
- Staff & Administration: manage user accounts, audit history and backups, subject to permissions.

For security, the application signs you out after 15 minutes without keyboard or mouse activity.

5. Customers and verification

- Open Customers and complete the new-customer form. Enter the customer's full name and phone number; add email and address if available.
- Search by customer name or ID. Select a result to review or update its details.

- For customer verification (KYC), choose a review status and add an internal note. The statuses are pending, verified and rejected.

New customers start as pending. Verification must be marked verified before an account can be opened or a loan application submitted. This application does not store identity-document images or numbers and does not perform real identity checks.

A customer can be deactivated only after all of their accounts are closed. Close accounts according to the account rules first.

6. Opening accounts and changing status

- In Accounts, find and select a customer. Choose Savings, Checking or Fixed Deposit.
- Enter the opening balance and create the account. Keep the account number available for later transactions.
- Select an account from the list to change its status. Freezing an account prevents its use; an authorized user can unfreeze it.

An account can be closed only when its balance is zero. A closed account cannot be reopened. Deposits and withdrawals are performed in Transactions.

7. Deposits, withdrawals and transfers

- Deposit: find the customer or account, select the account, enter the amount and an optional note, then submit the transaction.
- Withdrawal: select an account, enter the amount and submit. The amount cannot exceed the available balance.
- Transfer: choose separate source and destination accounts, enter the amount and submit. Both accounts must be active.
- Use History & Search to find past transactions by date, account or transaction reference.

Before submitting, verify the account, amount and destination. Completed transactions cannot be directly deleted or edited. If a correction is needed, ask an administrator to review the reversal process.

8. Loan applications and repayments

- A teller submits a loan request for a customer with a verified identity and active account. Enter the principal, annual percentage rate (APR) and term in months.
- An administrator reviews the request and approves or rejects it. Only an approved loan can be disbursed.
- A teller or administrator selects the approved loan and disburses it. The principal is deposited into the customer's selected active account. A teller records full or partial repayments.

Loan interest is estimated upfront using simple interest: principal × APR × term in years. This is an educational estimate, not an amortization schedule or a real credit decision.

9. Monthly account interest

- In Interest, select an account and preview the estimated credit. Review it before applying the interest to the account.
- Savings earns 2.00% APR and Fixed Deposit earns 3.50% APR. Checking accounts do not earn interest.

Interest is calculated against the balance at the time it is applied. It can be applied only once per account during the current UTC month.

10. Reports and exports

- Choose a report type, enter a start and end date in YYYY-MM-DD format, and select Generate.
- Transaction Summary counts deposits, withdrawals and transfers; Transaction Details lists individual transactions.
- Account Summary shows account types and statuses. Ledger Consistency Check compares stored balances with the transaction ledger.
- Loan Applications shows each request's status, principal, paid amount and outstanding balance; Loan Repayments lists individual payments. Interest Credits lists monthly credits by account.
- Leave a date blank for no date limit. Ledger Consistency Check and current balances in Account Summary ignore date limits.
- Choose Export CSV or Export PDF, select a location and save the file. CSV files can be opened in spreadsheet applications.

11. Staff, audit history and backups

- Administrators can create staff accounts, change roles, activate or deactivate accounts, and reset passwords.
- Audit Log records important changes. Search it by action, user or details. It is intended for review and accountability.
- Create a backup from the backup section. Store it in a secure location, and keep a trusted copy separate from this computer.
- To restore, select and confirm a backup file. The application first creates a safety copy of the current database. Restore signs you out; sign in again afterward.

Protect backup files as carefully as bank records. Do not restore an unknown or untrusted file.

12. Common issues

- A customer or account is missing: search again by name or account number and confirm the customer record exists.
- An account cannot be opened: confirm customer verification is marked verified and all required fields are complete.
- A transaction cannot be submitted: check your role, the account's active status and the available balance. Transfers require two different active accounts.
- A loan request cannot be submitted: customer verification must be complete and the selected account must be active.
- A report or export fails: confirm your report permission, correct date format and write access to the selected folder.
- You were signed out: this is expected after 15 minutes of inactivity. Sign in again.
- If the issue continues, contact an administrator and share the error message shown by the application.

Important reminder

This software is for training and practice only. Do not use it for real funds, sensitive real customer information or actual lending decisions.